

Notes: DeFusco, Johnson & Mondragon (RES, 2020)

Regulating Household Leverage

Contents

1	Big picture and incremental contribution	2
1.1	Incremental contribution	2
2	Data and measurement	2
2.1	Mortgage data and sample	2
2.2	Jumbo, conforming, DTI bins, and outcomes	2
3	Models and empirical specifications	3
3.1	Difference-in-differences pricing	3
3.2	Triple difference	3
3.3	Counterfactual DTI distribution and bunching	3
4	Identification and empirical design	3
5	Tables and figures	3
5.1	Table 1: Loan-level descriptive statistics	3
5.2	Figure 1: Mean interest rates by origination month for high-versus-low DTI jumbo loans . . .	4
5.3	Figure 2 and Table 2: price effects at the threshold	4
5.4	Figure 3: Flexible difference-in-differences estimates	5
5.5	Figures 4–7 and Table 3: quantity effects	6
5.6	Figure 8 and Table 4: product substitution	7
5.7	Figure 9 and Table 5: lender organization and agency frictions	8
5.8	Figure 10, Table 6, and Figure 11: default implications	9
6	Result map	9
7	Short summary	10
8	Conclusion	10
8.1	Core conclusion	10
8.2	Economic intuition	10
8.3	Incremental contribution	10
8.4	Policy implications	10
8.5	Limitations	10
8.6	Takeaway	10

1 Big picture and incremental contribution

- **Question.** How do mortgage credit markets respond when regulators impose an implicit cost on lenders that originate high-leverage household loans?
- **Policy setting.** The paper studies the Dodd–Frank Ability-to-Repay and Qualified Mortgage (ATR/QM) Rule, implemented in January 2014. The key threshold is a back-end debt-to-income ratio of 43%.
- **Main price result.** The rule increased interest rates on affected non-QM loans by roughly 10–15 basis points.
- **Main quantity result.** The policy eliminated about 15% of the affected high-DTI jumbo market and shifted about 20% of remaining high-DTI borrowers below the threshold.
- **Economic takeaway.** Macroprudential household leverage regulation can meaningfully reshape credit quantities even when price changes look small.

1.1 Incremental contribution

- The paper provides one of the first empirical evaluations of an ex ante U.S. household leverage regulation rather than ex post debt relief.
- It exploits the 43% DTI threshold and the jumbo/conforming exemption structure to estimate price and quantity effects.
- It develops a counterfactual DTI distribution using the exempt conforming market rather than assuming smooth bunching.
- It decomposes the response into interest-rate premia, bunching below the threshold, missing mass above the threshold, product substitution, lender-channel effects, and default implications.
- It shows that lender organization and agency frictions matter for regulatory incidence.

2 Data and measurement

2.1 Mortgage data and sample

The analysis uses loan-level mortgage data on origination month, loan amount, FICO score, LTV, DTI, interest rate, property characteristics, product type, origination channel, and investor type. The main sample focuses on first-lien conventional non-FHA 30-year fixed-rate purchase mortgages originated from 2010 to 2015.

Interpretation: The sample is designed to compare loans around the 43% regulatory threshold while keeping product and borrower characteristics comparable.

2.2 Jumbo, conforming, DTI bins, and outcomes

Jumbo loans are directly affected by the 43% DTI QM rule, while conforming loans are largely exempt through the GSE QM patch. DTI is rounded into one-percentage-point bins. The key outcomes are interest rates, DTI-bin originations, ARM shares, third-party origination shares, unknown-investor shares, and five-year defaults.

Interpretation: Jumbo loans supply the treated market; conforming loans help construct the untreated counterfactual.

3 Models and empirical specifications

3.1 Difference-in-differences pricing

$$r_i = \alpha + \beta_1 \mathbf{1}\{DTI_i > 43\} + \beta_2 \mathbf{1}\{DTI_i > 43\} \times Post_i + X_i' \gamma + \varepsilon_i.$$

Interpretation: β_2 is the post-ATR/QM premium for high-DTI loans.

3.2 Triple difference

$$r_i = \alpha + \beta [\mathbf{1}\{DTI_i > 43\} \times Jumbo_i \times Post_i] + controls + FE + \varepsilon_i.$$

Interpretation: The third difference uses exempt conforming loans as a comparison group.

3.3 Counterfactual DTI distribution and bunching

$$\begin{aligned} \hat{N}_d^{J,post} &= N_d^{J,pre} \times \frac{N_d^{C,post}}{N_d^{C,pre}}. \\ B &= \sum_{d=\bar{d}}^{43} (N_d^{emp} - \hat{N}_d^{cf}), \quad M = \sum_{d=44}^{50} (\hat{N}_d^{cf} - N_d^{emp}). \\ Intensive &= B/N_{44+}^{cf}, \quad Extensive = (M - B)/N_{44+}^{cf}. \end{aligned}$$

Interpretation: Bunching captures borrowers shifted below the threshold; missing mass minus bunching captures loans eliminated from the market.

4 Identification and empirical design

The paper combines threshold variation, before/after timing, and the jumbo/conforming exemption. Price effects are identified by changes around the 43% DTI threshold. Quantity effects are identified by comparing actual jumbo DTI distributions with counterfactual distributions built from the conforming market. Placebo years validate the counterfactual. Product substitution and lender-channel regressions diagnose mechanisms. Default calculations combine historical default gradients with the estimated distributional shift.

5 Tables and figures

5.1 Table 1: Loan-level descriptive statistics

Read: Jumbo loans are larger and have high FICO scores; the DTI-restricted sample remains comparable to the full sample except for higher DTI.

Detailed interpretation: The table establishes that the paper studies a prime conventional market, not merely a subprime tail.

Economic message: The regulation affects credit supply even in a high-quality borrower segment.

	Full sample			DTI $\in$ (36,50]		
	All loans	Conforming	Jumbo	All loans	Conforming	Jumbo
Loan amount (\$1,000's)	264.58 (189.75)	212.91 (101.26)	641.53 (249.59)	267.37 (187.31)	215.96 (101.23)	637.06 (240.84)
FICO score	756.12 (43.27)	754.50 (44.44)	767.91 (31.09)	751.09 (44.14)	749.03 (45.18)	765.93 (32.06)
Loan-to-value	80.34 (13.89)	80.81 (14.26)	76.96 (10.24)	80.51 (13.95)	81.06 (14.32)	76.55 (10.04)
Back-end debt-to-income	33.39 (9.04)	33.36 (9.10)	33.58 (8.61)	41.80 (3.51)	41.85 (3.52)	41.44 (3.39)
Interest rate	4.29 (0.56)	4.31 (0.56)	4.19 (0.54)	4.34 (0.54)	4.36 (0.54)	4.24 (0.54)
Number of observations	1,195,895	1,051,730	144,165	513,939	451,191	62,748

Notes:—This table presents loan-level descriptive statistics for both the full analysis sample (columns 1–3) and the restricted sample of loans with DTIs in a symmetric window around the QM-threshold of 43% used in the interest rate analysis (columns 4–6). The sample includes all first-lien, conventional (non-FHA), 30-year, fixed-rate, purchase mortgages originated between January 2010 and December 2015 for which CoreLogic reports a non-missing FICO, LTV, DTI, interest rate, appraisal amount and geographic identifier. All table entries represent sample means or, in parentheses standard deviations. Summary statistics are presented pooling across all loan types (columns 1 and 4) as well as separately

Table 1: Loan-level descriptive statistics

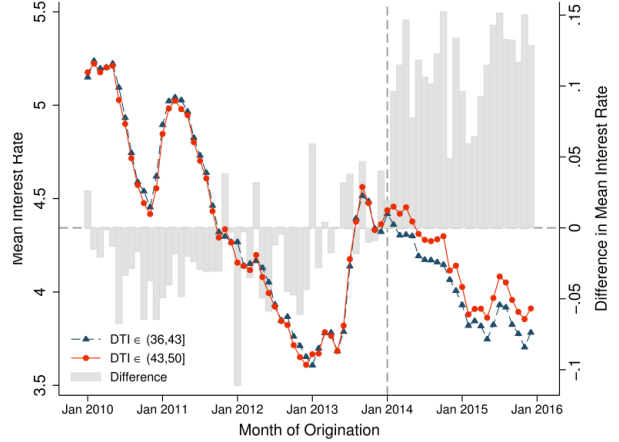


FIGURE 1
Mean interest rates by origination month for high-versus-low DTI jumbo loans
Notes:—This figure plots mean interest rates by origination month separately for high-DTI jumbo loans (circles) and low-DTI jumbo loans (triangles). Each dot represents the raw average interest rate for loans originated in the indicated month, measured on the left axis. The month-by-month difference in interest rates between high- and low-DTI loans is also plotted in grey bars and measured on the right axis. The vertical dashed line marks the month that the Ability-to-Repay Rule and Qualified Mortgage Standards went into effect (January 2014).

Figure 1: Mean interest rates by origination month

5.2 Figure 1: Mean interest rates by origination month for high-versus-low DTI jumbo loans

Read: Before 2014, high- and low-DTI jumbo rates move together; after ATR/QM, high-DTI rates rise relative to low-DTI rates.

Detailed interpretation: The timing supports the difference-in-differences design.

Economic message: Non-QM status increases prices, but the magnitude is modest.

5.3 Figure 2 and Table 2: price effects at the threshold

Read: Figure 2 shows a post-policy interest-rate jump above 43%; Table 2 estimates roughly 10–15 basis points of rate premium.

Detailed interpretation: The estimates remain stable across DiD and triple-difference specifications with rich controls.

Economic message: The price effect is threshold-based and regulatory in origin.

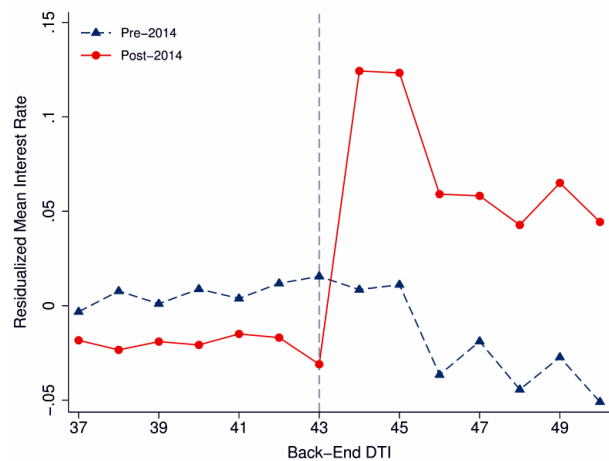


FIGURE 2
Detrended mean interest rates by DTI for jumbo loans originated pre- and post-ATR/QM
Notes:—This figure plots detrended mean interest rates by DTI separately for jumbo loans originated before (triangles) and after (circle).

(a) Figure 2

TABLE 2
The effect of non-qualified mortgage status on interest rates

	Difference-in-differences				Triple difference			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DTI > 43	-0.018*** (0.005)	-0.017*** (0.004)	-0.004 (0.004)	-0.009** (0.004)	0.003 (0.003)	0.003 (0.003)	0.019*** (0.002)	0.015*** (0.002)
DTI > 43 × Post	0.131*** (0.007)	0.141*** (0.008)	0.113*** (0.007)	0.115*** (0.007)	0.008*** (0.003)	0.008*** (0.003)	0.006*** (0.002)	0.008*** (0.002)
DTI > 43 × Jumbo					-0.021*** (0.006)	-0.022*** (0.006)	-0.032*** (0.005)	-0.030*** (0.005)
DTI > 43 × Jumbo × Post					0.124*** (0.008)	0.128*** (0.008)	0.108*** (0.007)	0.103*** (0.007)
Month (× Jumbo) FEs	X	X	X	X	X	X	X	X
County FEs		X	X	X		X	X	X
FICO × LTV FEs			X	X			X	X
Property Type FEs			X	X			X	X
FICO × LTV × Post FEs				X				X
Property Type × Post FEs				X				X
Implied %Δ	2.9%	3.2%	2.5%	2.6%	2.8%	2.9%	2.4%	2.3%
R-Squared	0.70	0.72	0.75	0.76	0.64	0.65	0.71	0.71
Number of observations	62,748	62,748	62,748	62,748	513,939	513,939	513,939	513,939

Notes:—This table reports difference-in-differences and triple-difference estimates of the effect of non-Qualified Mortgage status on interest rates. Each column reports a separate regression estimated at the loan level where the dependent variable is the interest rate (expressed in percentage points). Columns 1–4 report estimates from a difference-in-differences regression

(b) Table 2

5.4 Figure 3: Flexible difference-in-differences estimates

Read: Coefficients are near zero below 43% and jump above the regulatory threshold.

Detailed interpretation: This argues against a smooth risk-pricing explanation.

Economic message: Lenders price the legal boundary.

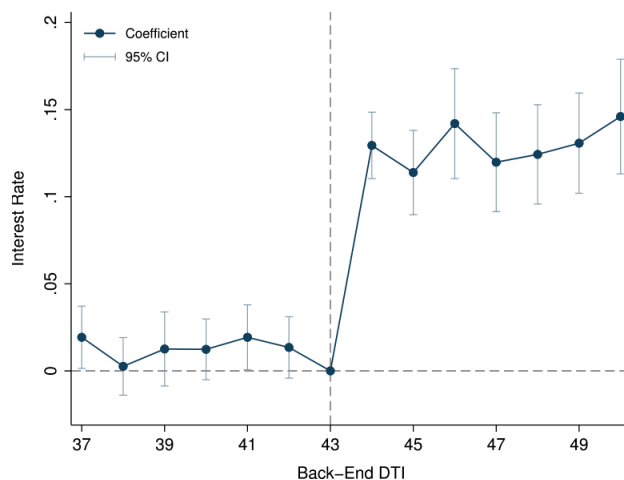


FIGURE 3
Flexible difference-in-differences estimates of the effect of non-QM status on interest rates
Notes:—This figure plots estimates of the effect of non-Qualified Mortgage status on interest rates derived from a flexible difference-in-differences specification that allows the effect to vary with the borrower's DTI. Estimates were constructed by regressing the interest rate on an indicator for whether the loan was originated after the implementation of ATR/QM and the interaction of that indicator with a series of dummies reflecting the borrower's DTI. The vertically dashed line marks the QM threshold of 43%. The DTI dummies

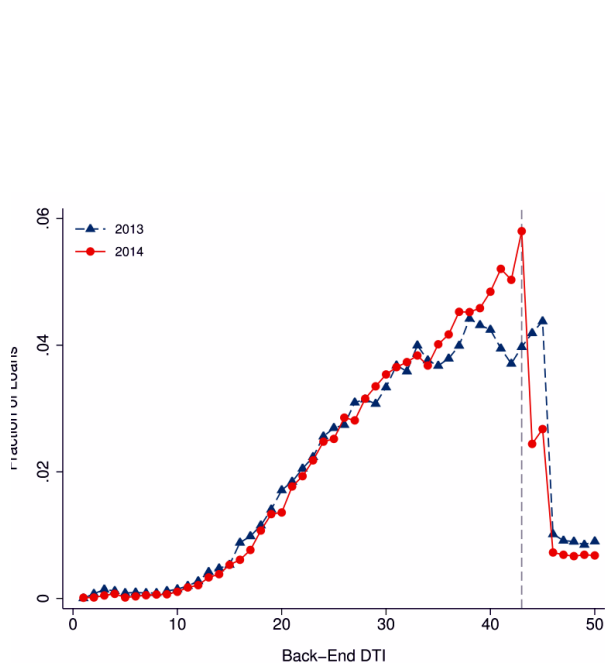
Figure 3: Flexible DiD estimates

5.5 Figures 4–7 and Table 3: quantity effects

Read: Figure 4 shows bunching below 43% and missing mass above it. Figure 5 supports the conforming-market counterfactual. Figure 6 validates the method in placebo years. Figure 7 estimates bunching and missing mass. Table 3 reports preferred intensive and extensive effects of 0.208 and 0.154.

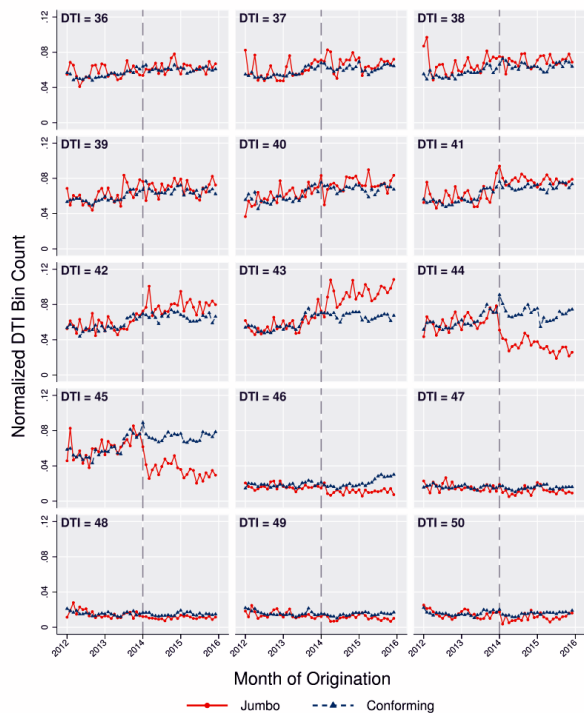
Detailed interpretation: These visuals and estimates show that the policy had a large quantity effect, separating borrowers who shifted DTI from loans that disappeared.

Economic message: ATR/QM rationed high-DTI jumbo credit rather than merely repricing it.

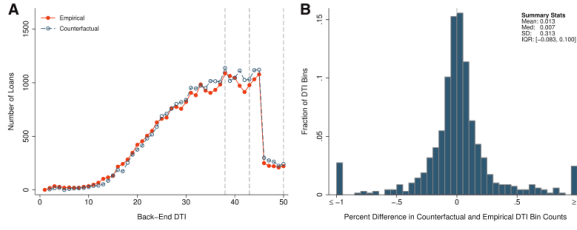


res.—This figure plots the distribution of DTI among jumbo mortgages separately for loans originated in 2013 (triangles) and 2014 (circles). Each dot represents the share of all mortgages originated in the indicated year for which the back-end DTI at origination fell

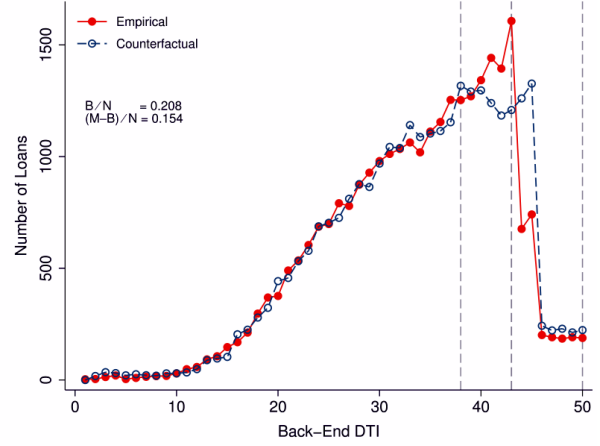
(a) Figure 4



(b) Figure 5



(a) Figure 6



(b) Figure 7

TABLE 3
Intensive and extensive margin effects of ATR/QM on the quantity of credit

	Preferred	Alternative specifications		
	(1)	(2)	(3)	(4)
	$\bar{d} = 38$	$\bar{d} = 30$	$\bar{d} = 35$	$\bar{d} = 40$
$\hat{B}/\hat{N}^{\text{post}}_{44+}$	0.208*** (0.033)	0.188*** (0.062)	0.269*** (0.048)	0.228*** (0.032)
$(\hat{M} - \hat{B})/\hat{N}^{\text{post}}_{44+}$	0.154*** (0.041)	0.180*** (0.069)	0.090 (0.056)	0.133*** (0.039)
Bootstrap replications	100	100	100	100
Number of observations	418,105	418,105	418,105	418,105

Notes:—This table reports estimates of the intensive and extensive margin effects of the Ability-to-Repay Rule and Qualified Mortgage standards on the quantity of credit in the jumbo mortgage market. The top row reports the estimated intensive margin effect of the regulation on the allocation of credit across the DTI distribution. Each estimate represents the fraction of jumbo loans in the counterfactual no-policy distribution that were shifted from a DTI above the QM-threshold of 43% to below the threshold. The second row reports the estimated extensive margin effect of the policy on the total number of jumbo mortgages originated. Each estimate represents the fraction of the counterfactual number of jumbo loans that were eliminated as a result of the policy. Intensive and extensive margin effects were calculated using the bunching procedure described in Section 5.1. Column one reports our preferred estimates, which set the lower limit of the bunching region to $\bar{d} = 38$. Columns 2–4 report analogous estimates from alternative specifications which set this limit to 30, 35, and 40%, respectively. All specifications use 2013 as the pre-period and 2014 as the post-period. The sample therefore includes all jumbo loans that were originated in either 2013 or 2014. Standard errors are reported in parentheses and are calculated by bootstrapping from the observed sample of mortgages, drawing 100 random samples with replacements and re-estimating

Table 3: Intensive and extensive effects

5.6 Figure 8 and Table 4: product substitution

Read: ARM shares rise near the threshold among jumbo loans; product-type bunching estimates show substitution is present but incomplete.

Detailed interpretation: Some borrowers reduce DTI by switching products, but many high-DTI loans are still eliminated.

Economic message: Product substitution is one margin of adjustment, not the whole effect.

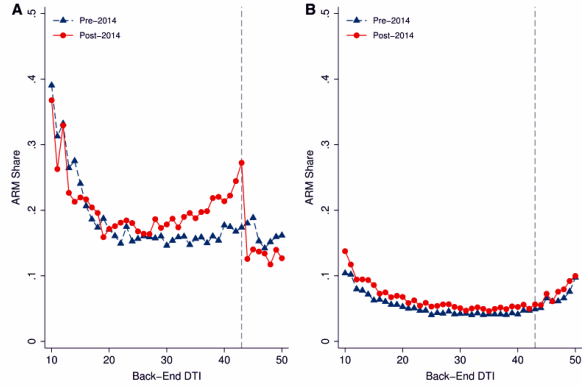


FIGURE 8
ARM share by DTI and market segment for loans originated pre- and post-ATR/QM
Notes:—This figure plots the share of mortgages with adjustable rates originated before (triangles) and after (circles) the implementation of ATR/QM separately by DTI and market segment. Panel A plots the ARM shares for jumbo mortgages only and panel B plots the analogous

(a) Figure 8

	FRMs Only	FRMs and ARMs Combined	ARMs Only
	(1)	(2)	(3)
$\hat{B}/\hat{N}_{44+}^{\text{post}}$	0.208*** (0.033)	0.333*** (0.032)	0.665*** (0.123)
$(\hat{M} - \hat{B})/\hat{N}_{44+}^{\text{post}}$	0.154*** (0.041)	0.101** (0.039)	-0.056 (0.146)
Bootstrap replications	100	100	100
Number of observations	418,105	454,360	36,255

Notes:—This table reports estimates of the intensive and extensive margin effects of the Ability-to-Repay Rule and Qualified Mortgage standards on the quantity of credit in the jumbo mortgage market across mortgage product types. Estimates are reported separately for FRMs (column 1), ARMs (column 3) and the pooled sample of fixed- and ARMs (column 2). The top row reports the estimated intensive margin effect of the regulation on the allocation of credit across the DT distribution. Each estimate represents the fraction of jumbo loans of the indicated type in the counterfactual no-policy distribution that were shifted from a DTI above the QM-threshold of 43% to below the threshold. The second row reports the estimated extensive margin effect of the policy on the total number of jumbo mortgages originated. Each estimate represents the fraction of the counterfactual number of jumbo loans of the indicated type that were eliminated as a result of the policy. Intensive and extensive margin effects were calculated using the bunching procedure described in Section 5.1 applied separately in each sample of loans. The lower limit of the bunching region is set to $\hat{d} = 38$ in all three samples. All specifications use 2013 as the pre-period and 2014 as the post-period. The sample therefore includes all jumbo loans of the indicated type that were originated in either 2013 or 2014. Standard errors are reported in parentheses and are calculated by bootstrapping from the observed sample of mortgages, drawing 100 random samples with replacement and estimating the parameter of each bootstrap. Coefficients levels 100%, 90%, and 10% are denoted by *, **, and ***.

(b) Table 4

5.7 Figure 9 and Table 5: lender organization and agency frictions

Read: Third-party and unknown-investor shares fall for high-DTI jumbo loans after ATR/QM.

Detailed interpretation: Lenders with more separation between origination, documentation, and risk-bearing retreat more.

Economic message: Agency costs help explain why quantities fall so much.

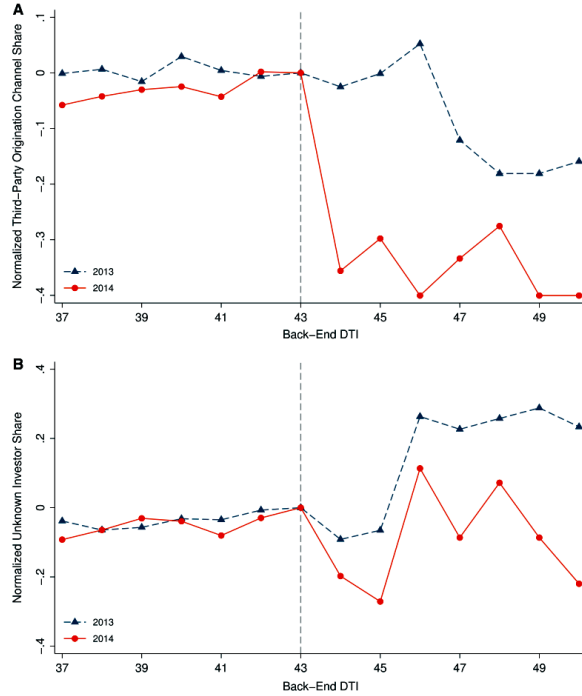


FIGURE 9
Third-party channel and unknown investor shares by DTI and origination year

(a) Figure 9

	Third-party channel		Unknown investor	
	(1)	(2)	(3)	(4)
DTI > 43	-0.009 (0.009)	-0.006 (0.009)	0.012 (0.012)	-0.000 (0.010)
DTI > 43 × Post	-0.311*** (0.023)	-0.298*** (0.022)	-0.045 (0.036)	-0.051* (0.029)
Month FEs	X	X	X	X
County FEs	X	X	X	X
FICO × LTV FEs	X	X	X	X
Property type FEs	X	X	X	X
FICO × LTV × Post FEs	X	X	X	X
Property type × Post FEs	X	X	X	X
Number of observations	22,685	22,685	22,685	22,685

Notes:—This table reports difference-in-differences estimates of the effect of non-Qualified Mortgage status on the likelihood that a loan is originated through a third-party channel (columns 1–2) or held by an unknown investor (column 3–4). Each column reports a separate regression estimated at the loan level in the sample of jumbo loans with DTI between 36 and 50% with non-missing origination channel and investor status. Third-party originations include all loans originated through the correspondent, broker, or wholesale channels. Unknown investors include only instances in which the data explicitly indicates that the investor was unknown rather than being a portfolio investor (*i.e.* loans with missing investor status are not included). Investor status is measured in the third month after origination to avoid misclassifying loans that are temporarily held in portfolio before being sold to unknown, non-portfolio investors. Coefficient estimates are reported for the non-QM “treatment” dummy (DTI > 43) as well as its interaction with an indicator for whether the loan was originated in a month following the implementation of ATR/QM (Post). Columns 1 and 3 include fixed effects for the month of origination. Columns 2 and 4 add fixed effects for the county the property is located in, the borrower’s FICO score (20-point bins), LTV (5-percent bins), and property type (single family, condominium, townhouse, planner).

(b) Table 5

5.8 Figure 10, Table 6, and Figure 11: default implications

Read: The jumbo DTI-default relationship is weak in the affected region; aggregate default effects are modest.

Detailed interpretation: The rule reduces high-DTI lending, but the targeted loans are not sufficiently more likely to default to imply a large direct default-rate reduction.

Economic message: The policy's financial-stability benefits may come from channels beyond individual DTI-default risk.

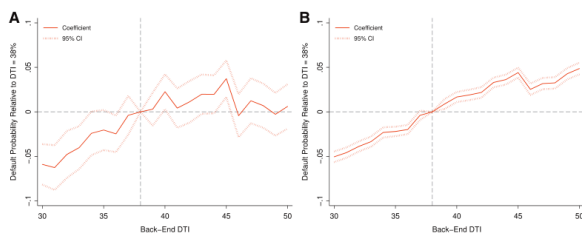


FIGURE 10
Relationship between DTI and 5-year default probability (2005-8)

Notes:—This figure plots the empirical relationship between DTI at origination and the probability of default for loans originated during 2005-8. Panel A is constructed using a sample of jumbo mortgages only, whereas Panel B is based on a sample of both jumbo and non-jumbo mortgages. Each panel reports the coefficient estimates from a regression of whether a loan defaulted on a series of dummy variables indicating whether the loan's DTI fell into a given one-percent bin. We define a loan as having defaulted if the borrower was ever more than 90 days delinquent or if the property was repossessed by the lender (foreclosure or REO) within 5 years of the origination date. We omit the dummy for DTI < 38, which is the lower limit of the binning region in our preferred specification for the quantity effect. DTI bins are created by rounding up to the nearest integer so that the 38% bin includes all DTIs greater than 37% and less than or equal to 38%. The regressions also included fixed effects for the month of origination, the county the property was located in, the type of property as well as 20-point FICO score bins, 5-point LTV bins and the pairwise interaction between the two. The 95% confidence intervals are based on standard errors that were clustered at the county level.

(a) Figure 10

TABLE 6
Estimates of the effect of DTI on the 5-year probability of default

	(1)	(2)	(3)	(4)
	2005	2006	2007	2008
Panel A: Jumbo loans only				
DTI ≤ 38	-0.0303*** (0.0060)	-0.0555*** (0.0105)	-0.0910*** (0.0086)	0.0025 (0.0381)
DTI > 43	0.0034 (0.0062)	-0.0112 (0.0081)	0.0002 (0.0082)	0.0779** (0.0365)
Implied aggregate effect	-0.0009* (0.0004)	-0.0002 (0.0007)	-0.0020** (0.0009)	-0.0053** (0.0022)
High-DTI bin share	0.22	0.25	0.27	0.20
Medium-DTI bin share	0.22	0.24	0.23	0.19
Number of observations	31,529	18,646	17,155	1,186
Panel B: All loans				
DTI ≤ 38	-0.0330*** (0.0018)	-0.0508*** (0.0026)	-0.0689*** (0.0028)	-0.0706** (0.0038)
DTI > 43	0.0062*** (0.0022)	0.0083*** (0.0025)	0.0228*** (0.0029)	0.0320** (0.0044)
Implied aggregate effect	-0.0010*** (0.0002)	-0.0017*** (0.0003)	-0.0035*** (0.0005)	-0.0040** (0.0006)
High-DTI bin share	0.19	0.23	0.26	0.23
Medium-DTI bin share	0.20	0.22	0.22	0.19
Number of observations	353,392	330,550	295,674	91,493

Notes:—This table reports estimates of 5-year default probabilities for high-DTI and low-DTI loans relative to loans in the omitted category DTI ∈ (38, 43]. The relationship between DTI and default probability is estimated separately by origination year cohort and loan type. Panel A reports results for jumbo loans only whereas Panel B pools across all loans. The third row of each panel also reports the implied counterfactual effect of the ATP/CAL rule on the aggregate

(b) Table 6

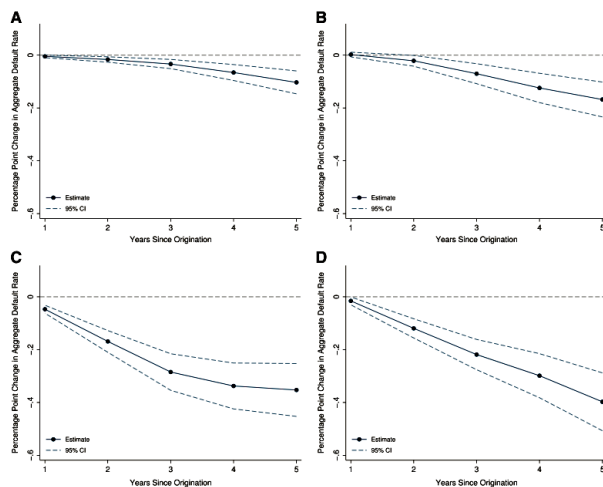


FIGURE 11

Figure 11: Aggregate default-rate effects

6 Result map

- Price pass-through is moderate: about 10–15 basis points.

- Quantity effects are large: roughly 20% shifted below the threshold and 15% eliminated.
- The custom counterfactual is validated in placebo years.
- Product substitution explains part, but not all, of the effect.
- Lender organization matters: third-party and unknown-investor channels retreat.
- Direct default-rate reductions are small relative to the credit contraction.

7 Short summary

ATR/QM reduced household leverage by changing the DTI distribution, but much of the adjustment came through credit supply contraction rather than rate increases. The paper’s central empirical lesson is that macroprudential rules can have large quantity effects in intermediary markets with agency and documentation frictions.

8 Conclusion

8.1 Core conclusion

ATR/QM raised the cost of non-QM jumbo loans modestly but produced a large decline in high-DTI credit. The rule successfully reduced leverage, but partly by restricting credit availability.

8.2 Economic intuition

A legal safe-harbor rule changes lender incentives. When documentation, litigation, and secondary-market frictions are important, lenders may exit the regulated segment instead of simply charging higher rates.

8.3 Incremental contribution

The paper contributes by providing one of the clearest empirical evaluations of ex ante household leverage regulation, by developing a counterfactual bunching method for nonsmooth distributions, and by showing that market organization shapes regulatory incidence.

8.4 Policy implications

Policymakers should measure both prices and quantities. A small price change can hide a large credit contraction. Exemptions and lender business models are central to the incidence of leverage regulation.

8.5 Limitations

The paper does not observe lender identities, does not fully model general-equilibrium effects on house prices, and estimates default implications using historical pre-crisis relationships.

8.6 Takeaway

Regulating household leverage is powerful but blunt. It reduces risky leverage, but its welfare effects depend on how lenders respond, which borrowers lose access, and whether the targeted loans would actually have generated systemic losses.